Smart Door Lock System Using Arduino

Realised by : ZAKARI Boutayna

CodeAlpha 2026

Introduction

- Security is essential in homes, offices, and institutions
- Traditional locks can be duplicated or broken easily
- Digital systems provide better access control

Image Source

Problem Statement

- Conventional locks lack advanced protection
- Unauthorized access can compromise sensitive areas
- Need for affordable smart security solutions

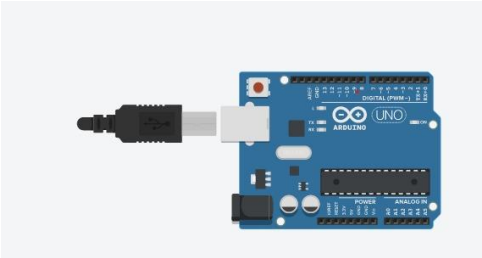
Proposed Solution

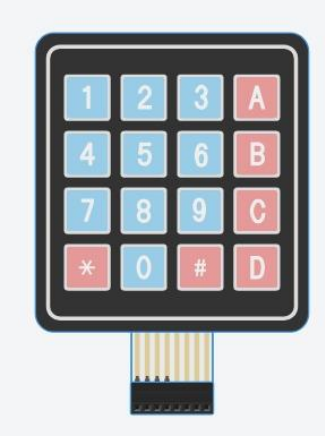
- Password-based electronic door locking system
- Arduino processes user authentication
- Servo motor controls door lock mechanism

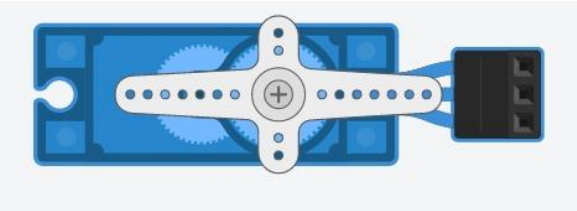
LIST OF COMPONENTS

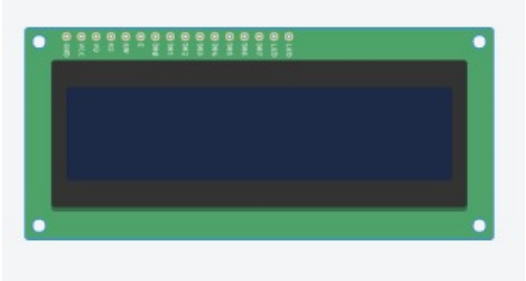
1. Arduino uno r3
2. LCD 16*2
3. 1k ohm Resistor
4. 220 ohm Resistor
5. Keypad 4*4

6.Positional Micro Servo







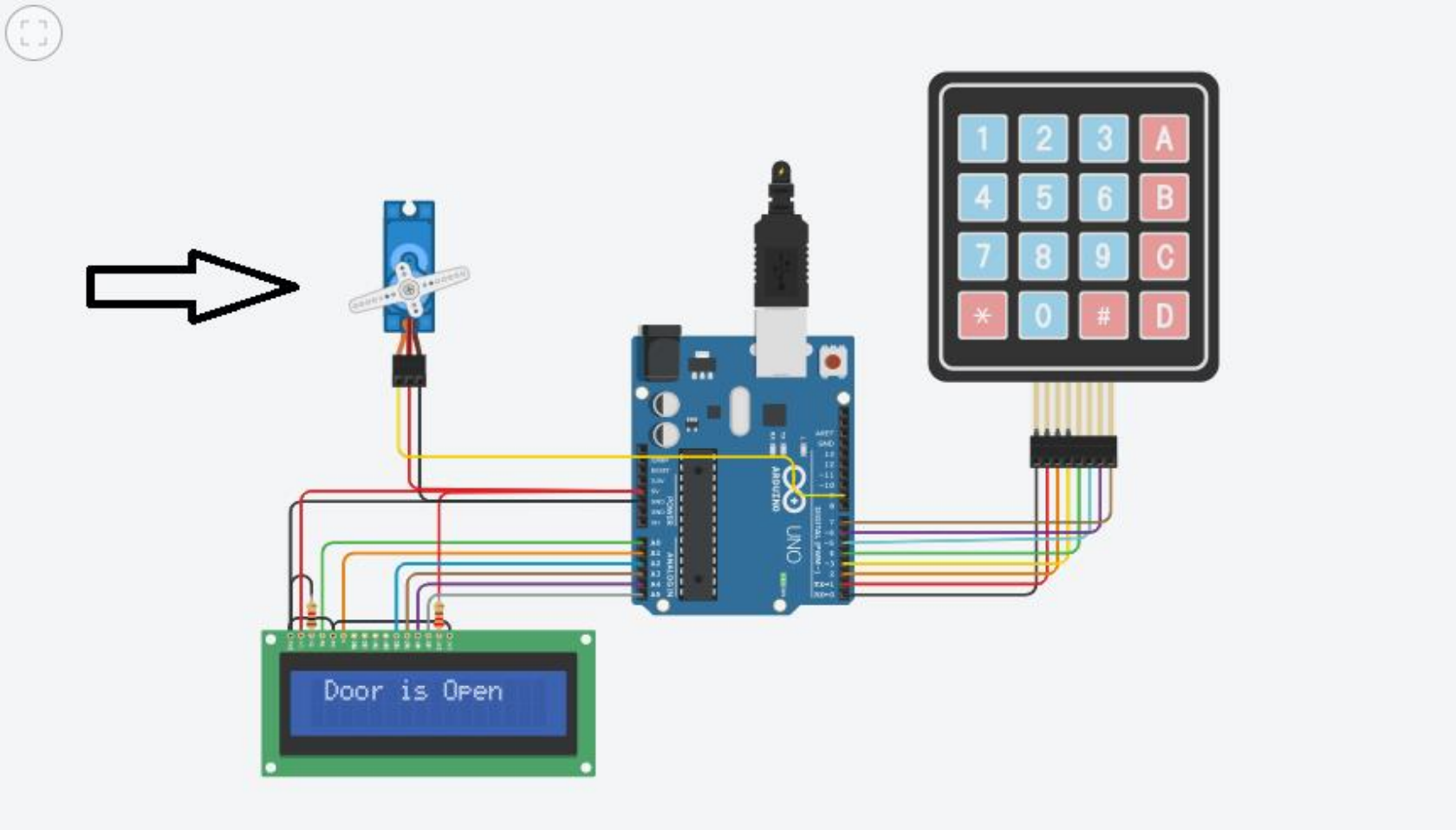


Software

Tinker cad online website

https://www.tinkercad.com/

CIRCUIT DIAGRAM

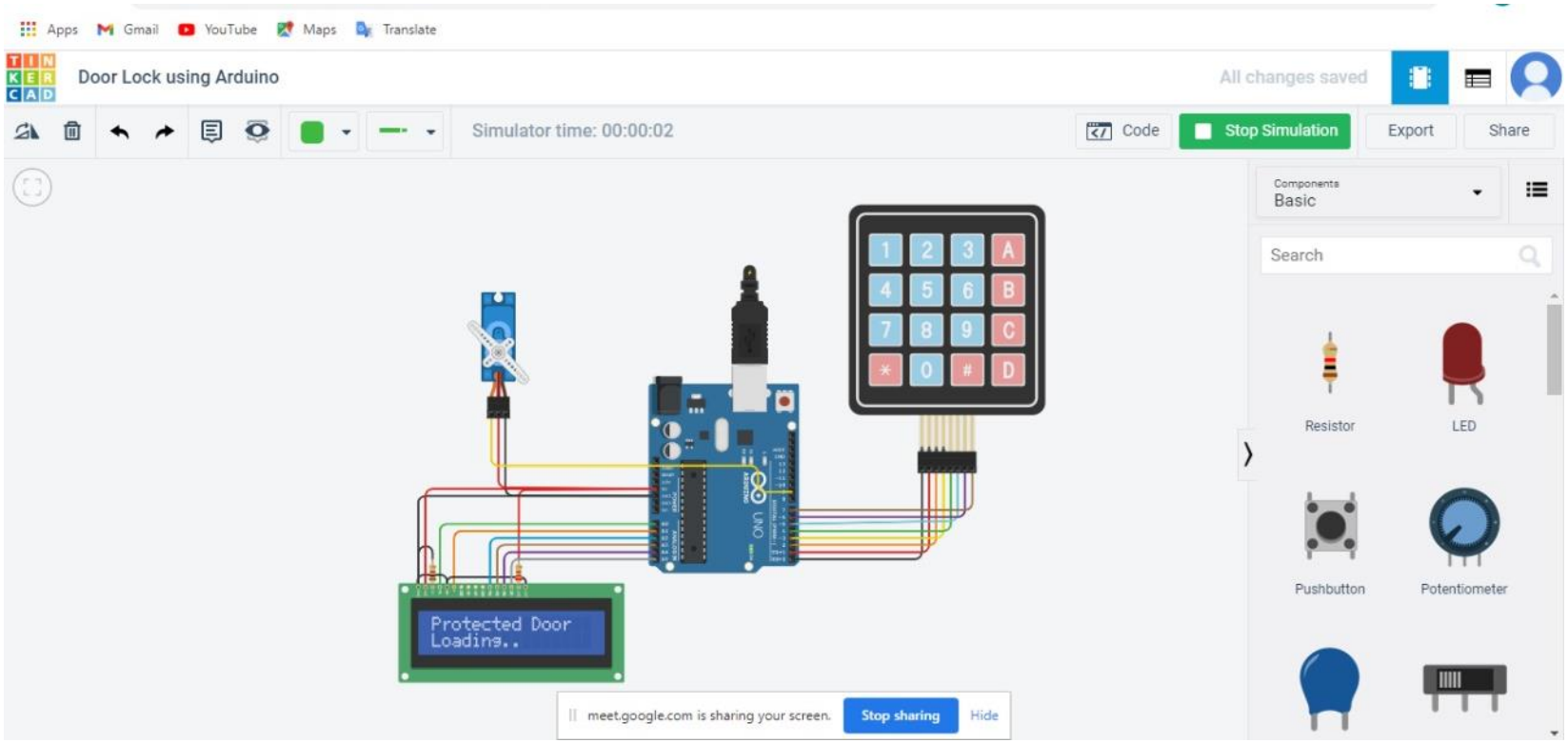


CIRCUIT DIAGRAM OF DOOR OCK SYSTEM

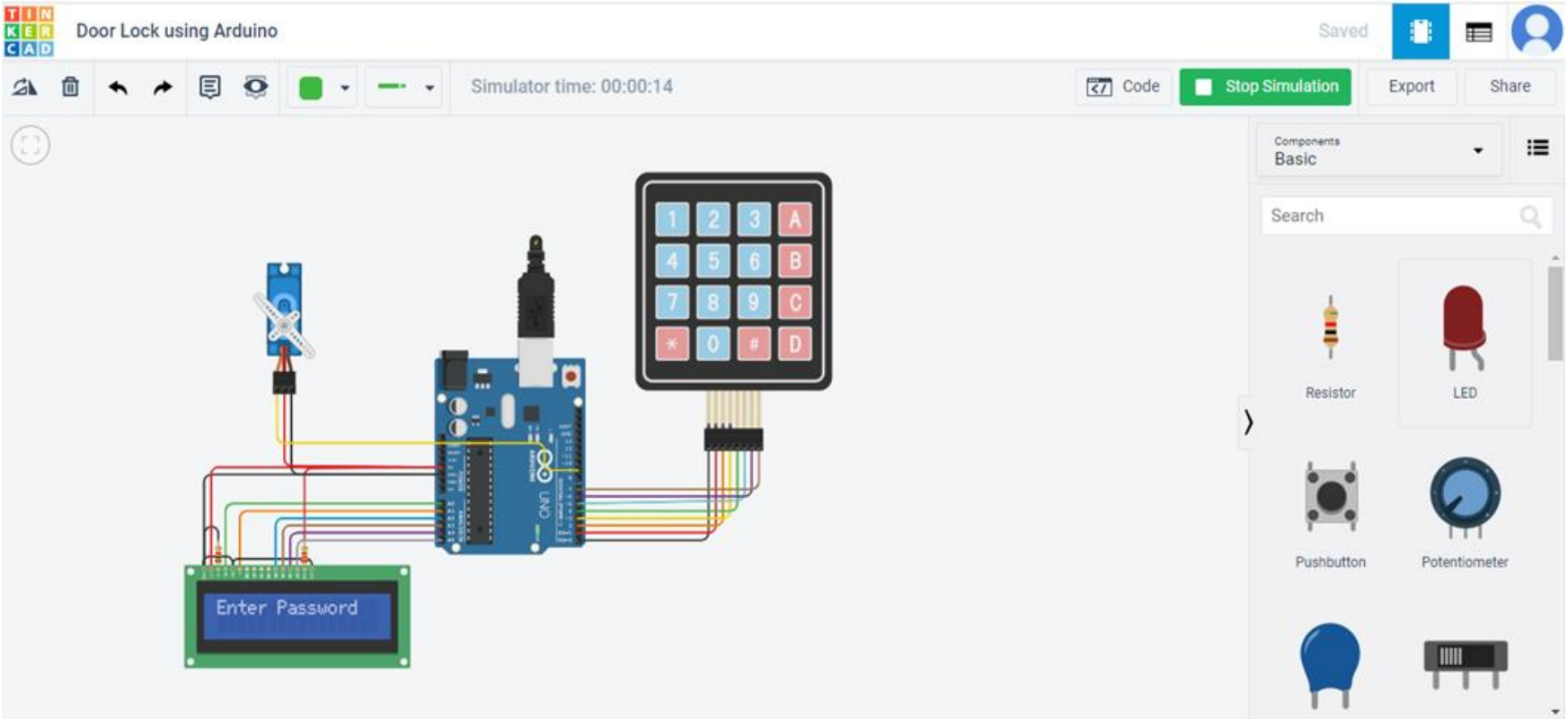
APPLICATIONS

- Password based door lock system is used widely in places where we need to secure something
- It is used in Banks and in places where some valuable information has to be secured
- It can also be used in place of normal door locks as they are more secure & reliable

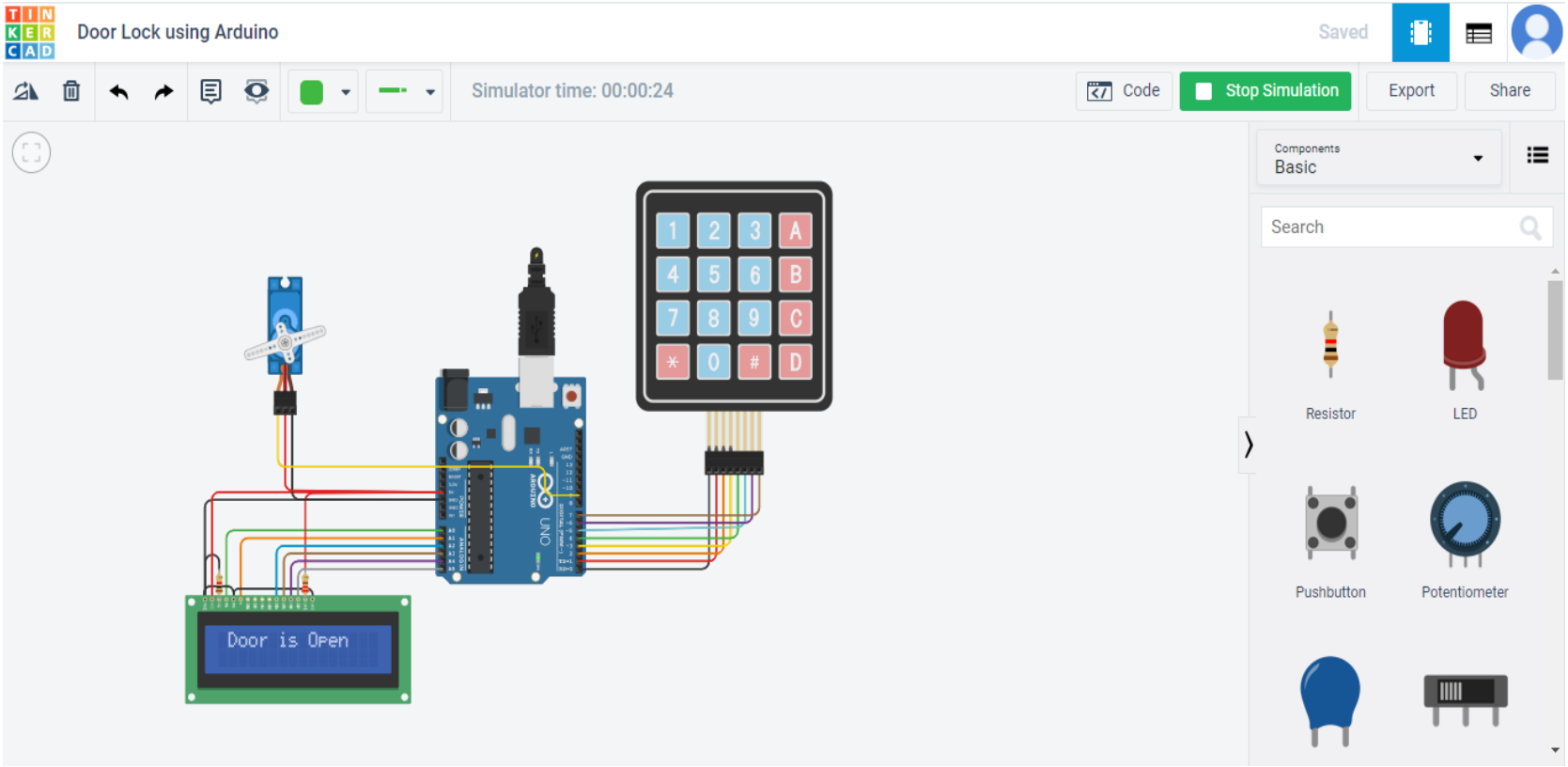
RESULT(SIMULATION)



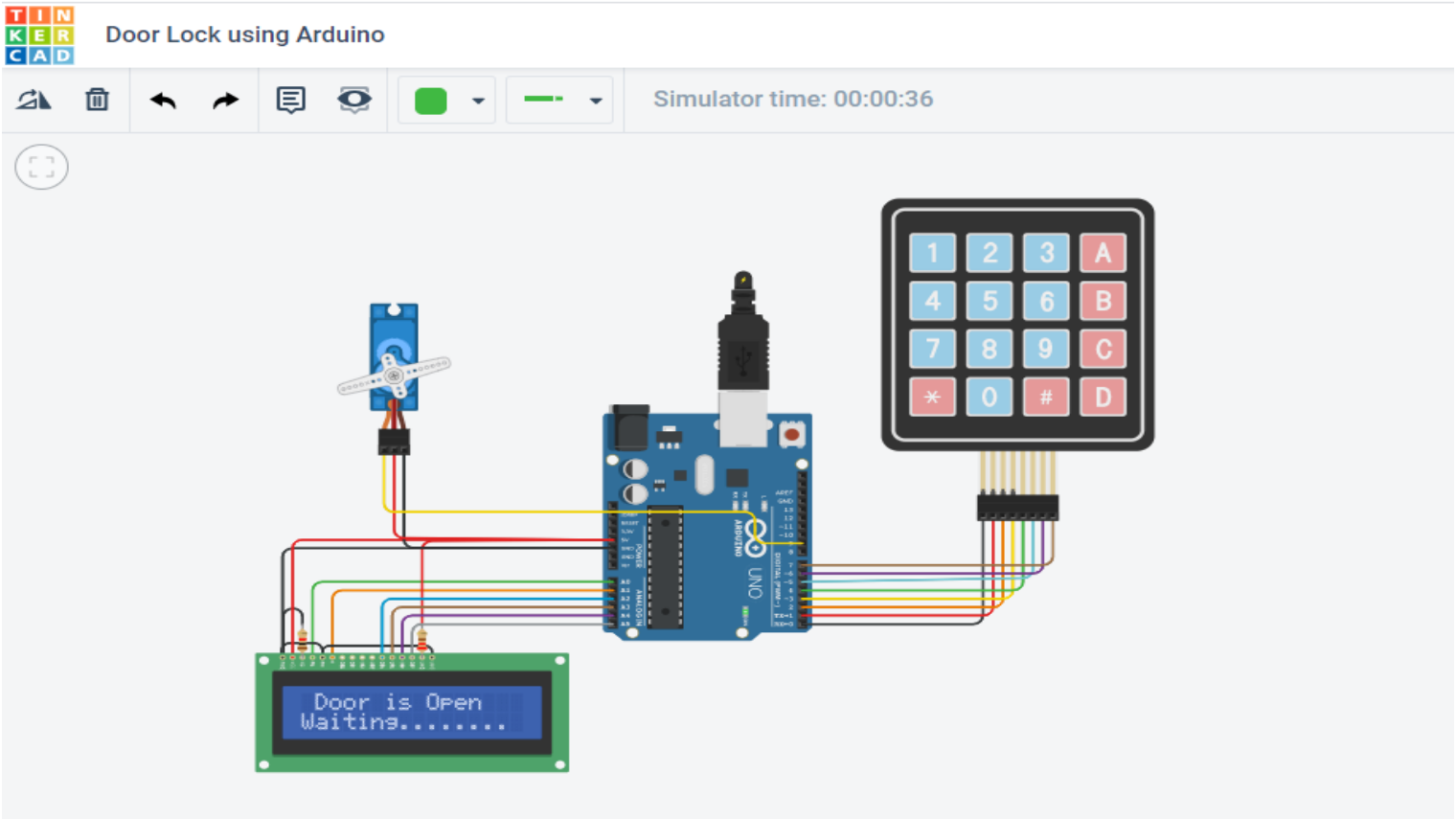
CASE 1: Setup of door lock system using Arduino



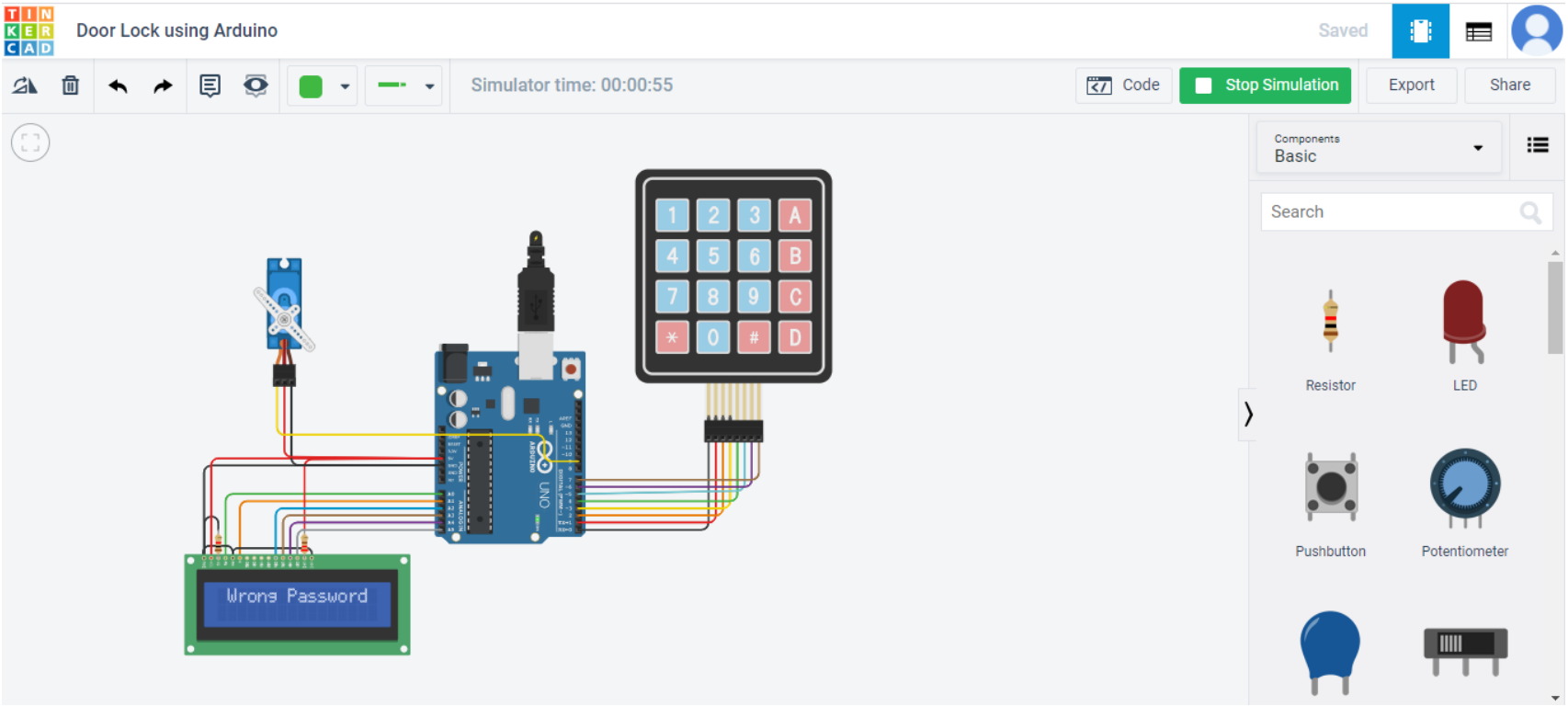
<u>CASE 2</u>:Working of door lock system when enter password is shown after setup



<u>Case 3</u>: Working of door lock system after the password is entered



<u>CASE4:</u>When the door is waiting for action after after entering the password



<u>CASE5</u>:Working of Door lock system when wrong password is entered

CONCLUSION

- Provides a low-cost smart security solution
- Easy to implement using Arduino technology
- Can be upgraded with RFID, fingerprint, or IoT